Aspen Johnson

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Education

Valdosta State University
B.S in Data Science: Computational Sci. & Eng. Track, Minors in Physics and Math — *Expected: 2028, GPA: N/A*

Palm Beach State College | Academic transition year
A.A in STEM: Mechanical/ Aerospace Engineering Track — *Aug 2025 - May 2026, GPA: 3.2*

Mercer University
Coursework in Finance, Mathematics, and Economics — *June 2022 - May 2025, GPA: 3.0*

Publications & Research Presentations

An empirical application of improved gradient scaling for score-driven volatility filters — Published

- Selected as 1 of 15 research papers to present at the 2026 Atlanta Research Conference.
- Georgia Undergraduate Research Conference (GURC), November 2026 – Accepted for presentation.

Score-Driven Dynamic Fractional Integration with Heavy Tails: An Application to Earthquake Activity — Under Review

Research Experience

Undergraduate Research | Long-Memory Time Series & Seismicity Modelling — June 2026 – October 2026
Faculty Guided Research w/ Prof. Szabolcs Blaszek — *Remote*

- Independently proposed and initiated a research investigation into earthquake dynamics with the aim of explooring the science behind seismic activity.
- Developed a Python-based data acquisition workflow using USGS earthquake catalog queries, compiling and analyzing over two million seismic observations spanning more than 50 years across California and Alaska.
- Investigated earthquake-count series, inter-event intervals, and seismic activity clustering as potential measures of temporal dependence, with an emphasis on **long-memory behavior** and **score-driven fractional integration models**.

Undergraduate Research | Computational Physics & Molecular Dynamics — April 2026 – Present
Faculty Guided Research w/ Prof. Shafat Mubin — *On-Site/Remote*

- Studied the **Velocity Autocorrelation Function (VACF)** to analyze how particle velocities evolve over time in molecular dynamics trajectories.
- Developing familiarity with **LAMMPS simulation workflows**, including **interatomic potentials**, **time integration**, **thermostats**, and **atomic trajectory analysis**.

Undergraduate Research | Score-Driven Volatility Models — Jan 2025 – Dec 2025
Faculty Guided Research w/ Prof. Szabolcs Blaszek and Adam Kobor — *On-Site/Remote*

- Contributed to work showing that Hessian-driven gradient scaling can improve volatility-filter performance relative to existing scaling methods.
- Gained publication experience through faculty-guided research involving model interpretation, literature review, empirical analysis, and manuscript development.
- Analyzed financial time-series data across Bitcoin, S&P 500, Gold, U.S. REITs, and CHF/JPY from 2010–2025 using R, GAUSS, Excel, and Bloomberg Terminal data.

Industry & Work Experience

Financial Analyst Intern — May 2025 – Dec 2025
Zinzino USA — *Remote*

- Reduced chargeback processing time by 87% through workflow digitization and documentation streamlining.
- Refined the charge back process within a few weeks of arrival, and curated a reference guide to aid future onboarding employees.
- Created clear documentation that enabled cross-departmental collaboration and made Finance operations easy for shareholders and upline leaders to understand.

Projects

CourtSignal | ***Assistive Technology for Hearing Impaired Athletes*** In Progress

Independent Research & Engineering Project

- Developing an assistive technology concept leveraging visual and haptic signaling to improve real-time communication accessibility in competitive athletic environments.
- Selected for the University System of Georgia and Georgia Tech's inaugural 2026 Startup Studio seminar, supporting customer discovery, entrepreneurship, and early-stage product validation.

Custom Cyberdeck In Progress

Independent Engineering Project | *Portable, Powerful, and Playable*

- Learning the theory behind designing a compact, handheld computing system optimized for extended use during collegiate athletic travel, prioritizing ergonomics, portability, and battery efficiency.

Leadership, Athletics, and Service

Collegiate Women's Basketball Player June 2022 – Present

Mercer University | *Valdosta State University* *NCAA Division I* | *Division II*

- Commit 20–25+ hours per week to practices, strength training, film study, travel, competition, recovery, and team responsibilities while balancing full-time academic and research commitments.
- Develop leadership, discipline, time management, communication, resilience, and performance under pressure through year-round collegiate athletics.
- Represent university athletic programs in competitive, team-based environments requiring accountability, preparation, adaptability, and consistent execution.

Leadership, Community Engagement & Extracurriculars

Read2Succeed | *Tutoring elementary students to aid in building literary in Macon, Georgia*

Putnam Mathematics Competition Prep | *Independent Study*

Youth Sports Volunteer | *Assisted young athletes K-12 in a variety of sports.*

Weekly Physics Study Group Member | *Going through **Helliwell's Modern Classical Mechanics** text*

Book Club Fellow | *Selected as one of ten students to participate through Mercer Universities Economics Department*

Content Creation | *Live streamed my workflow on YouTube amassing ∼20,000 views*

AWS Campus Prep Series Participant | *Completed a Summer-long series of webinars regarding Amazon Careers and Cloud Services*

Technical Skills

Languages in order of proficiency: Python, Java, LaTeX, HTML/CSS, R, Gauss, C/C++, JavaScript, Octave, TypeScript, JavaScript

Developer Tools: Git, CLI, VSCode

Other

Clubs/Organizations:NSBE, SWE, and Codecademy's Women in Tech

Relevant Coursework: Calculus I–III, Linear Algebra, Differential Equations (ODE), Probability and Statistics, Data Structures, Programming, Intro Econometrics, Time Series Analysis in R, Physics I